

Lakshya JEE 2.0 2025

Maths

DPP: 1

Basics Mathematics

Q1

- (a) If $\log_{10}(x^2 - 12x + 36) = 2$
 (b) $9^{1+\log x} - 3^{1+\log x} - 210 = 0$; where base of log is 3.

Q2 If $\frac{x}{7} = \frac{y}{11} = \frac{z}{13} = \frac{2x-3y+z}{m}$, then the value of m is

- (A) 6
 (B) -6
 (C) $\frac{1}{6}$
 (D) $-\frac{1}{6}$

Q3 Solve $x : \log^2(4-x) + \log(4-x) \cdot \log(x + \frac{1}{2})$ for

$$-2\log^2\left(x + \frac{1}{2}\right) = 0.$$

Q4 Values of: $\log_{(\sqrt{3}-\sqrt{2})}(5 - 2\sqrt{6})$ is

- (A) -2
 (B) $\frac{1}{2}$
 (C) 2
 (D) $-\frac{1}{2}$

Q5 The set of all solutions of the inequality

$$\left(\frac{1}{2}\right)^{x^2-2x} < \frac{1}{4} \text{ contains the set}$$

- (A) $(-\infty, 0)$
 (B) $(-\infty, 1)$
 (C) $(1, \infty)$
 (D) $(3, \infty)$

Q6 If $\log_x(\log_{18}(\sqrt{2} + \sqrt{8})) = \frac{1}{3}$, Then the value of $1000x$ is equal to

- (A) 8 (B) $1/8$

(C) $1/125$

(D) 125

Q7 Simplify:

$$5^{\log_{1/5}(\frac{1}{2})} + \log_{\sqrt{2}} \frac{4}{\sqrt{7}+\sqrt{3}} + \log_{1/2} \frac{1}{10+2\sqrt{21}}$$

Q8 Value of: $\log_3(\log_2 512)$ is equal to

- (A) $\frac{1}{2}$
 (B) 2
 (C) -2
 (D) $-\frac{1}{2}$

Q9 Simplified value of expression:

$$\log_6 18 + \log_6 2 - \log_2 3 - \log_2 \left(\frac{1}{3}\right) =$$

- (A) 2 (B) 1
 (C) 0 (D) -1

Q10 Complete set of values of x for which $\log_3(|x| - 5)$ is defined is

- (A) $x \in (-\infty, -5] \cup [5, \infty)$
 (B) $x \in (5, \infty)$
 (C) $x \in (-5, 5)$
 (D) $x \in (-\infty, -5) \cup (5, \infty)$

Q11 $\frac{1}{1+\log_b a + \log_b c} + \frac{1}{1+\log_c a + \log_c b} + \frac{1}{1+\log_a b + \log_a c}$, has the value of equal to

- (A) abc
 (B) $\frac{1}{abc}$
 (C) 0
 (D) 1

Q12 Number of real roots of an equation

$$3^{\log_3(x^2-4x+3)} = x - 3 \text{ is}$$

- (A) 0 (B) 1
 (C) 2 (D) Infinite



Q13 The sum of all positive integral value of 'a' $a \in [1, 500]$ for which the equation $[x]^3 + x - a = 0$ has solution ([.] represents greatest integer function)

- (A) 462 (B) 512
(C) 784 (D) 812

Q14 Number of non-negative integral values of x satisfying the inequality

$$\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \geq 0$$

- (A) 0 (B) 1
(C) 2 (D) 3

Q15 The complete solution set of the inequality

$$\frac{x^4-3x^3+2x^2}{x^2-x-30} \geq 0 \text{ is:}$$

- (A) $(-\infty, -5) \cup (1, 2) \cup (6, \infty) \cup \{0\}$
(B) $(-\infty, -5) \cup [1, 2] \cup (6, \infty) \cup \{0\}$
(C) $(-\infty, -5) \cup [1, 2] \cup [6, \infty) \cup \{0\}$
(D) $(-\infty, -5] \cup [1, 2] \cup [6, \infty)$

Q16 The number of the integral solutions of

$$x^2 + 9 < (x + 3)^2 < 8x + 25 \text{ is:}$$

- (A) 1 (B) 3
(C) 4 (D) 5

Q17 Calculate:

- (a) $7^{\log_3 5} + 3^{\log_5 7} - 5^{\log_3 7} - 7^{\log_5 3}$
(b) $4^{5 \log_{4\sqrt{2}}(3-\sqrt{6}) - 6 \log_8(\sqrt{3}-\sqrt{2})}$

Q18 If $x + y = a$ and $x^2 + y^2 = b$, then value of $(x^3 + y^3)$, is

- (A) ab
(B) $a^2 + b$
(C) $a + b^2$
(D) $\frac{3ab-a^3}{2}$

Q19 $\left(\frac{5}{6}\right)^{3/4}$ when divided by $\left(\frac{5}{6}\right)^{7/6}$ becomes

- $\left(\frac{5}{6}\right)^{7-x}$ the value of x is
(A) $\frac{7}{12}$
(B) $\frac{89}{12}$

- (C) $\frac{8}{12}$
(D) $\frac{10}{12}$

Q20 $(2d^2e^{-1})^3 \times \left(\frac{d^3}{e}\right)^{-2} =$

- (A) $8e^{-2}$
(B) $8e^{-3}$
(C) $8e^{-1}$
(D) $8e^{-4}$

Q21 Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set $C = \{f : A \rightarrow B \mid 2 \in f(A) \text{ and } f \text{ is not one-one}\}$ is

Q22 If $\frac{(2^{n+1})^m \cdot 2^{2n} \cdot 2^n}{(2^{m+1})^n \cdot 2^{2m}} = 1$ then m is equal to

- (A) 0 (B) 1
(C) n (D) $2n$



Answer Key

Q1 (a) 16, -4
(b) 5

Q2 (B)

Q3 $x = 0, 7/4, \frac{3+\sqrt{24}}{2}$

Q4 (C)

Q5 (D)

Q6 (D)

Q7 6

Q8 (B)

Q9 (A)

Q10 (D)

Q11 (D)

Q12 (A)

Q13 (D)

Q14 (D)

Q15 (B)

Q16 (D)

Q17 (a) 0
(b) 9

Q18 (D)

Q19 (B)

Q20 (C)

Q21 19

Q22 (D)



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